AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 9

Sustainability Concept & Strategy

Prepared by: Assim
4th-Year Student, AI & Data Science — aivancity
July 2026

Consolidates sustainability-relevant decisions already made and verified across Deliverables 1, 4, and 7.

9.1 Applicable Standards — Precisely Scoped

[CONFIRMED] Dubai's official green building framework, Al Sa'fat, is building-envelope focused (thermal insulation, U-values, glazing, HVAC, renewable energy) and is mandatory for any structure requiring a Dubai Municipality building permit. In this masterplan, that applies specifically and only to the Restroom/Comfort Station — the one permitted building in the design. It does not directly govern open landscape, paving, or planting design, which has no equivalent mandatory certification found in this research.

Practical implication: the Restroom/Comfort Station should target Al Sa'fat Silver at minimum (the mandatory baseline for all new Dubai buildings) — external walls ≤0.57 W/m²K, roof/ceiling ≤0.30 W/m²K, glazing ≤2.0 W/m²K with SHGC ≤0.25. These are detailed-design engineering values, noted here so they aren't lost between this concept stage and construction documentation.

[HONEST GAP] No landscape-specific Dubai sustainability certification exists for the park itself. Rather than invent one or cite an inapplicable building standard for open space, this strategy is built from Dubai Municipality's own stated current practices (recycled-water irrigation, heat-mitigating material choices) — sourced, not certified.

9.2 Climate-Responsive Design — Already Built In, Not Bolted On

This is the one area where sustainability wasn't treated as a separate late-stage layer — it shaped the masterplan from Deliverable 1 onward. Recap:

Confirmed edge condition	Design response
NW edge: harshest — summer sunset (296° azimuth) + prevailing shamal wind	Buffer Grove: dense Ghaf/Samar windbreak planting, no primary gathering space sited here
SE edge: mildest — winter morning sun, wind-sheltered	Arrival Commons: main entrance sited here by design, not convenience
SW edge: mosque-adjacent within 89.0m	Quiet Turf Buffer + graduated transition zone before active courts — an acoustic/activity response, not just a solar one
Site-wide: 40-49°C summer highs, 9-10.5 sunshine hrs/day	70.8-73.7% verified shade coverage over both playgrounds (Deliverable 4); shade canopy rhythm along the full promenade

Full sourcing and solar-geometry calculation: Deliverables 1 and 5.

9.3 Water Strategy

[CONFIRMED] Irrigation drawn from recycled/treated water rather than potable supply — Dubai Municipality's own stated practice for its landscaping projects (Deliverable 7), adopted here rather than left as an unresolved tension.

Element	Water strategy
Buffer Grove, Promenade, Arrival planting (Ghaf, Sidr, Neem, Samar)	All confirmed low-water, DM-planted desert-adapted species — not generic drought-tolerant selections (Deliverable 7)
Event Lawn (Central Green Commons)	Real irrigated turf, not artificial — the brief's open-lawn requirement genuinely needs living grass for both comfort and DM's own program precedent; this is the highest-water element in the design and is stated as such, not hidden

Element	Water strategy
Primary paths & plazas	Permeable PICP paving reduces runoff and supports groundwater recharge rather than shedding stormwater off-site
Perimeter/buffer beds	Light gravel mulch reduces evaporation loss versus bare soil

[OPEN ITEM] No specific irrigation volume or water budget has been calculated — that requires a landscape/irrigation engineer working from final planting density and local evapotranspiration data, outside this project's scope. Flagged as a real next step, not silently assumed to be fine.

9.4 Materials & Heat Mitigation

Every hardscape material decision in Deliverable 7 was made primarily for heat performance, not aesthetics — restated here as sustainability content since urban heat island mitigation is a core sustainability criterion:

- **Light-toned PICP/natural stone paving** throughout — high solar reflectance versus dark asphalt/pavers
- **Light-tone EPDM playground surfacing** specifically to reduce surface temperature versus standard dark rubber
- **No artificial turf anywhere** (reaches 60-80°C in direct Dubai sun) — a genuine safety and sustainability constraint, not just a materials preference

[CONFIRMED] 62 of 74 existing trees (84%) are retained in the current masterplan — preserving established canopy and root systems rather than starting the site's ecological value from zero. The 12 flagged for likely removal are point-based conflicts with placed hard elements, not a blanket clearance (Deliverables 3-4).

9.5 Energy & Lighting

[CONFIRMED] Evening lighting strategy (visually confirmed in Deliverable 6 renders) uses warm bollard and canopy-integrated lighting along the accessible path loop, not stadium-style floodlighting — floodlighting is reserved specifically for the Padel/Basketball courts, where it's a functional requirement for evening play, not general park illumination.

[OPPORTUNITY — NOT YET COMMITTED] The shade sail canopy structures (5 along the promenade, 2 dedicated playground canopies) present a real opportunity for integrated solar PV, given their orientation and Dubai's consistent sun exposure. This is flagged as a genuine option for the engineering/detailed-design phase, not claimed as a current design feature — no structural or electrical load calculation has been done to confirm feasibility.

9.6 Biodiversity & Ecology

- **Ghaf (Prosopis cineraria),** the UAE's national tree and a keystone desert species, anchors the Buffer Grove — supports local birds and pollinators beyond its role as a windbreak
- **84% tree retention** (9.4) preserves an existing, already-established micro-habitat rather than resetting it
- **The Buffer Grove's dense windbreak planting** doubles as a habitat edge along the site's least-programmed boundary — an ecological function layered onto an already climate-driven design decision, not a separate add-on

9.7 Operational Sustainability

[CARRIED FORWARD FROM DELIVERABLE 4] The confirmed 170m cart-haul distance for waste/maintenance (no vehicle access to the NW Service/Utility Gate) has a real operational-sustainability dimension: more

frequent, longer manual servicing trips versus a vehicle-accessible bin store. Still an open item for the operations/FM conversation, not resolved by this document.

Material durability was also selected with maintenance load in mind: synthetic sports surfaces and EPDM playground surfacing are UV-stable products by design, appropriate for Dubai's sun exposure rather than requiring frequent replacement.

9.8 Summary — What's Resolved vs. What's Still Open

Resolved	Open
Climate-responsive zoning (edge-specific, not generic)	Specific irrigation water budget
70.8-73.7% verified playground shade compliance	Solar PV feasibility on shade structures
Recycled-water irrigation strategy adopted	170m cart-haul operational cost — flagged, not fixed
Heat-mitigating material palette throughout	Local/regional material sourcing not yet confirmed
84% existing tree retention	Restroom building's specific Al Sa'fat compliance path

---

Sources: Deliverables 1, 4, 5, 7 (this project); Al Sa'fat — Dubai Green Building System (Dubai Municipality, 2nd Edition); Dubai Municipality public landscaping and irrigation practice. Full AI-tool-by-tool methodology log: Deliverable 10 — AI Methodology Report.